Data Example: Poverty in the United States

Spatial Regression Models

Mahedi Hasan and Jun Zhu

Exploratory Data Analysis

In this demonstration, we will explore the American Community Survey data across the continental United States.

We first load the libraries we will be using, `spdep`, `tidyr`, `ggplot2`, `gridExtra`, `rgdal`, `MASS`, `sp`, and `ncf`.

```
library(spdep)
library(spatialreg)
library(tidyr)
library(ggplot2)
library(gridExtra)
library(sf)
library(MASS)
library(sp)
library(ncf)
```

Background

The American Community Survey is an ongoing statistical survey conducted by the U.S. Census Bureau and the data are publicly available.

Our research objectives are to quantify the spatial pattern of county-level poverty rates in the continental United States and to assess the relationship between county-level poverty rates and several socio-economic factors.

We focus on data for the census year 2000 in the 48 states of the continental United States. The original data at the individual level are aggregated to the county level. The counties are identified by the 5-digit federal information processing (FIP) standards code, with the first two digits corresponding to a state code and the last three digits corresponding to a county code.

The poverty rate (`povty`) is defined to be a percentage of individuals between the ages of 18 and 64 in a county who lived in poverty in the census year 2000. The socio-economic factors are:

- `ag`: % work in agricultural sector
- `manu`: % work in manufacture sector
- `retail`: % work in retail sector
- `foreign`: % foreign born
- `feemp`: % female employment
- `hsch`: % completed high school education
- `black`: % black
- `hisp`: % Hispanic

We next load the data set into a data frame called `povdf` using the R function `read.table()`. To include variable names, we specify `header = TRUE`. Since we are loading a comma-separated file (`csv`), we also specify

sep = ",".

```
setwd("E:\\Postdoctoral Research Associate - UW Madison\\Teaching\\2026 Spring STAT679\\Data")

povdf <- read.table("uspoverty2000.csv", header = TRUE, sep = ",")
```

We check the number of rows in the data frame `povdf` using the function `nrow()`.

```
nrow(povdf)
```

```
## [1] 3070
```

We explore the structure of the data frame `povdf` using the `str()` function.

```
str(povdf)
```

```
## 'data.frame':    3070 obs. of  12 variables:
##  $ FIPS_N  : int  1001 1003 1005 1007 1009 1011 1013 1015 1017 1019 ...
##  $ ag      : num  2.13 1.57 3.97 3.3 2.18 ...
##  $ black   : num  17.11 10.29 46.32 22.2 1.19 ...
##  $ feemp   : num  53.2 50.5 43.5 44.3 48.1 ...
##  $ foreign : num  1.17 2.11 1.5 0.43 3.1 ...
##  $ hisp    : num  1.4 1.76 1.65 1.01 5.33 ...
##  $ hsch    : num  33.8 29.6 32.4 35.7 36 ...
##  $ manu    : num  16.5 12.5 31.4 23.8 19.5 ...
##  $ povty   : num  9.06 9.13 21.91 17.74 9.89 ...
##  $ retail  : num  12.85 14.2 10.51 9.93 11.61 ...
##  $ STATEFP : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ COUNTYFP: int  1 3 5 7 9 11 13 15 17 19 ...
```

The data frame `povdf` features the poverty rate and the socio-economic factors from the census year 2000 and in the counties of the 48 continental states. Using the R function `head()`, we view the first 6 counties in the data frame `povdf`.

```
head(povdf)
```

```
##   FIPS_N    ag black feemp foreign hisp  hsch  manu povty retail STATEFP
## 1   1001  2.13 17.11 53.21    1.17 1.40 33.83 16.48  9.06  12.85       1
## 2   1003  1.57 10.29 50.51    2.11 1.76 29.61 12.54  9.13  14.20       1
## 3   1005  3.97 46.32 43.52    1.50 1.65 32.41 31.37 21.91  10.51       1
## 4   1007  3.30 22.20 44.32    0.43 1.01 35.73 23.75 17.74   9.93       1
## 5   1009  2.18  1.19 48.10    3.10 5.33 36.01 19.52  9.89  11.61       1
## 6   1011 10.46 73.11 36.54    3.06 2.75 35.23 23.22 28.60   7.53       1
##   COUNTYFP
## 1        1
## 2        3
## 3        5
## 4        7
## 5        9
## 6       11
```

To perform exploratory data analysis, we use summary statistics and graphical methods.

Using the R function `summary()`, we obtain several univariate summary statistics for each of the variables in the data frame `povdf`.

```
summary(povdf)
```

```
##      FIPS_N            ag              black            feemp
##  Min.   : 1001   Min.   : 0.030   Min.   : 0.000   Min.   :23.21
```

```
##  1st Qu.:19032   1st Qu.: 1.680   1st Qu.: 0.290   1st Qu.:46.94
##  Median :29178   Median : 3.790   Median : 1.675   Median :51.88
##  Mean   :30457   Mean   : 6.129   Mean   : 8.710   Mean   :51.64
##  3rd Qu.:45047   3rd Qu.: 7.810   3rd Qu.: 9.865   3rd Qu.:56.30
##  Max.   :56045   Max.   :55.600   Max.   :86.490   Max.   :78.47
##     foreign           hisp             hsch            manu
##  Min.   : 0.000   Min.   : 0.080   Min.   :10.93   Min.   : 0.000
##  1st Qu.: 0.890   1st Qu.: 0.910   1st Qu.:30.86   1st Qu.: 8.945
##  Median : 1.710   Median : 1.780   Median :34.91   Median :14.965
##  Mean   : 3.419   Mean   : 6.230   Mean   :34.80   Mean   :15.938
##  3rd Qu.: 3.900   3rd Qu.: 5.107   3rd Qu.:38.99   3rd Qu.:21.997
##  Max.   :50.940   Max.   :97.540   Max.   :53.25   Max.   :48.550
##      povty           retail         STATEFP         COUNTYFP
##  Min.   : 2.04   Min.   : 1.72   Min.   : 1.00   Min.   :  1.00
##  1st Qu.: 8.51   1st Qu.:10.30   1st Qu.:19.00   1st Qu.: 35.00
##  Median :11.61   Median :11.56   Median :29.00   Median : 77.00
##  Mean   :12.69   Mean   :11.48   Mean   :30.36   Mean   : 97.14
##  3rd Qu.:15.70   3rd Qu.:12.72   3rd Qu.:45.00   3rd Qu.:131.00
##  Max.   :53.83   Max.   :26.90   Max.   :56.00   Max.   :810.00
```

For a continuous variable, the summary statistics are its minimum, first quartile, median, mean, third quartile, and maximum. The summary statistics of the response variable of poverty rate show that the lowest county-level poverty rate was 2.04% and the highest county-level poverty rate was 53.83% among the 3070 counties in the continental United States in 2000. The center of the county-level poverty rates is 11.61% measured by median and 12.69% measured by mean (i.e., average). In addition, the interquartile range is between the first quartile of 8.51% and the third quartile of 15.70%. That is, half of the counties had a poverty rate below 11.61% and the other half had a poverty rate above 11.61%, while the middle half of the counties had a poverty rate between 8.51% and 15.70%. Among the explanatory variables, we take the female employment rate for example. The summary statistics of the female employment rate tell us that the county-level female employment rate ranged between the lowest 23.21% and the highest 78.47% among the 3070 counties in the continental United States in 2000. Half of the counties had a female employment rate below the median of 51.88% and the other half had a poverty rate above 51.88%, while the middle 50% of the counties had a female employment rate between 46.94% and 56.30%.

Using the R function `cor()`, we obtain the sample correlation, a bivariate summary statistic, between poverty rate and any given socio-economic factor. For example, the correlation between poverty rate and female employment rate is in the output below.

```
cor(povdf$povty, povdf$feemp)
```

```
## [1] -0.6900635
```

The sign of this sample correlation is negative, indicating a negative correlation between the poverty and female employment rates. That is, higher female employment rates are associated with lower poverty rates, whereas lower female employment rates are associated with higher poverty rates. The magnitude of this correlation is 0.69 (rounded from 0.6900635) and reflects a moderate amount of association between the poverty and female employment rates for such an observational study in the social sciences.

Visualization

We explore the relationship between the poverty rate and socio-economic factors by using the `ggplot2` package to create a histogram and scatter plot.

We use the function `ggplot()` to create a density histogram of `povty`. We use the function `geom_freqpoly()` and specify the number of bins to be 100 by setting `bins = 100`. We create a title with `ggtitle()`, and omit the grey background with `theme_bw()`. We save this histogram as object `p1`.

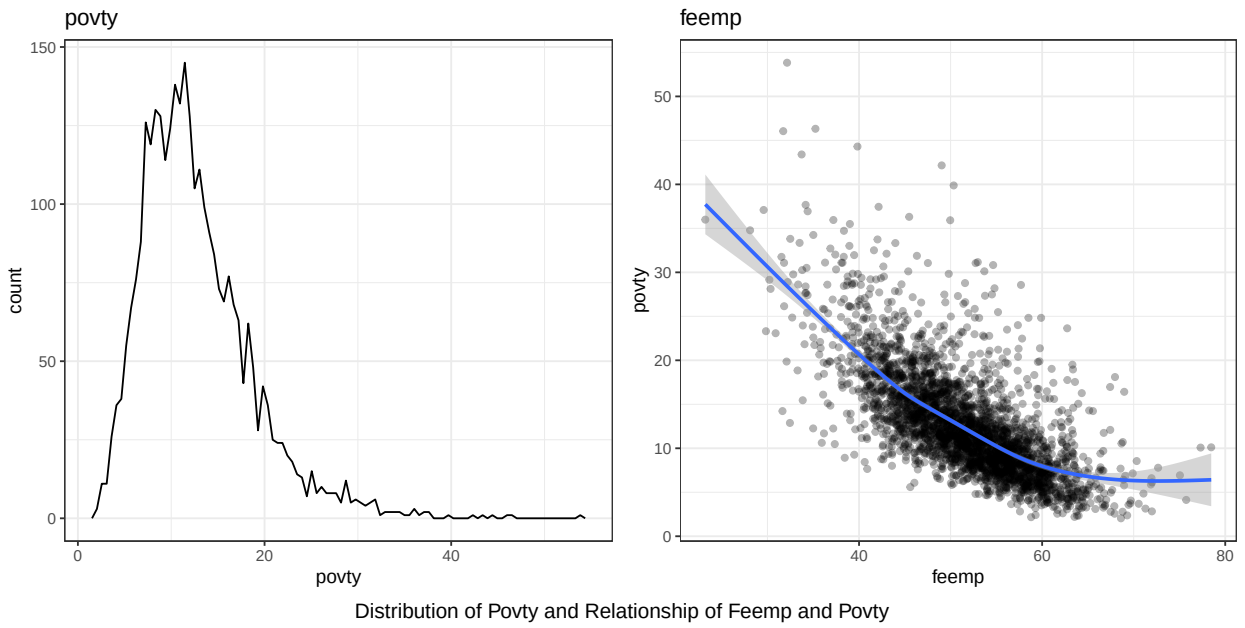
To create the scatter plot, we use the `ggplot()` function and plot `feemp` on the x-axis and `povty` on the y-axis. We create the points using the function `geom_point()`. We add additional features such as specifying the opacity of the points with the argument `alpha` and setting it to `0.3`. We also add a localized smoothing line with the addition of `geom_smooth()`. We save this scatter plot as object `p2`.

```
p1 <- ggplot(povdf, aes(x = povty)) +
  geom_freqpoly(bins = 100) +
  ggtitle("povty") +
  theme_bw()

p2 <- ggplot(povdf, aes(x = feemp, y = povty)) +
  geom_point(alpha = 0.3) +
  geom_smooth() +
  ggtitle("feemp") +
  theme_bw()
```

Having created the histogram and the scatter plot and stored them as objects `p1` and `p2`, we combine them into one figure using the function `grid.arrange()` from the package `gridExtra`. We enter the histogram and scatter plot to combine, `p1` and `p2`, and specify the number of columns the figure will have using the argument `ncol = 2`. We add an overall title to the bottom of the plot using the optional argument, `bottom =`.

```
grid.arrange(
  p1, p2, ncol = 2,
  bottom = "Distribution of Povty and Relationship of Feemp and Povty"
)
```



The histogram shows that the range of poverty rates is between 0 and 55% with a center of just above 10%. The histogram is also right skewed revealing counties with high poverty rates in the right tail. The histogram is based on density such that the areas of the vertical bars at 10% increments add up to a total probability of one. Alternatively, one could plot the histogram by the number of counties at the 10% increments and the shape of the histogram would be the same.

The scatter plot shows a negative trend. As the female employment rate increases from 20% to nearly 80%, the poverty rate declines. This finding is consistent with the negative sample correlation, indicating a negative association between female employment rate and poverty rate.